Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I need switch housing

Assistant: To help you select the best material for your switch housing, could you please clarify:

1. What is the exact use of the switch housing? For example, is it for a wall switch, appliance, industrial equipment, railway system, or another specific application?
2. Where will this switch housing be used in your solution? Is it for consumer electronics, industrial controls, home appliances, or another end-use?

Your answers will help me understand the requirements and operating environment for the housing.

User: consumer electronics indoor and main body

Assistant: Thank you for clarifying. To further narrow down the requirements for your indoor consumer electronics switch housing (main body), could you please provide:

1. What are the typical operating conditions? For example, will the housing be exposed to elevated temperatures, humidity, or cleaning agents, or is it mostly in a stable indoor environment?
2. What is the minimum wall thickness of your part?
3. Are there any preferred base materials you would like to use (such as PC, ABS, PBT, etc.), or are you open to suggestions?

4. If you have a preferred material, are you open to alternative material suggestions, or do you need to stick with your specified base material?

These details will help ensure the material options are suitable for your application.

User: need thickness of 0.8 mm and no preference

Assistant: Here are the key material requirements (CTQs) assumed for your switch housing application:

1. Minimum wall thickness: 0.8 mm
2. Flame rating: UL 94 V-0 at 0.8 mm
3. Comparative Tracking Index (CTI): ≥ 350 V
4. Glow Wire Flammability Index (GWFI): 960°C
5. Glow Wire Ignition Temperature (GWIT): 750°C
6. Processing method: Injection molding
7. Flame retardant feature: Halogen-free preferred

Are all of these requirements correct for your application? Please confirm or let me know if any need adjustment.

User: this looks good you can recommend the grades

Assistant: Here are potential options for your switch housing application, all meeting your requirements for indoor consumer electronics, 0.8 mm wall thickness, and critical electrical safety features:

1. CELANEX® XFR 6842 GF30
2. CELANEX® XFR 6852 GF30
3. Zytel® FR70G25V0GWNH NC010

4. Crastin® FR684NH1 NC010

5. FRIANYL® A3 H GF25 V0 NC 1102/EA

6. Zytel® HTNFR42G30NH NC010

Each grade is halogen-free, flame retardant, and suitable for injection molding. They offer strong electrical insulation, glow wire compliance, and are proven in similar switch housing applications. Please review these options for fit with your design and manufacturing needs. Final material selection and product design decisions rest with you, and suitability should be validated for your specific requirements. If your application involves critical safety or regulatory compliance, ensure all applicable standards are met, as Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.